Detection and isolation of fault from cyber attack in SCADA systems using network parameters

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Abstract. The SCADA system is a vital system that monitors and controls industrial processes. In this system, a cyber attack occurs by infiltrating the communication channels between sensors, actuators and servers. In recent years, cyber attacks have caused problems for industrial control systems. In terms of disrupting the system, a cyber attack acts similarly to the occurrence of a fault in it. Having instantaneous access to network parameters such as end-to-end delay, packet loss and network traffic can make it possible to tell a fault from a cyber attack. The purpose of this research is first to detect and then to isolate faults from cyber attacks in the SCADA system by using network parameters. For this purpose, a fluid-flow system together with a controller has been modeled. A Beard fault detection filter (BFDF) has been designed for this system, which identifies the various anomalies of the system. If, at the time the filter detects an anomaly, the network parameters also show abnormal conditions, a cyber attack is detected. The simulations were carried out in Omnet++ software. The results of the research showed that this method is effective in separating faults from cyber attacks.

Keywords: Fault detection, cyber attack, DoS attack, SCADA system, Beard fault detection filter (BFDF), OMNET++ software.

1. Introduction

The SCADA system is a type of cyber-physical system (CPS) used in the critical infrastructure of countries, such as the power grid, urban water supply networks, oil refineries and chemical plants. The use of communication networks in these systems has made them vulnerable to cyber attacks. Possible cyber attacks on these systems may have a negative effect on the economy, the environment and national security [1, 2]. Among the latest cyber attacks on the country's facilities, one can name the attack on the smart fuel system in Aban 1400 (October–November 2021).

In general, the physical part of a CPS is monitored or controlled by its cyber layer, which includes sensing, computation and communication. However, the cyber layer is vulnerable to cyber attacks and can indirectly affect the physical system. Cyber attacks can be divided into three main categories: the denial-of-service (DoS) attack, the replay attack and the false data injection (FDI) attack. A DoS attack can disrupt communication channels so that the reception of data is prevented [3–5]. A replay attack records and covers the communication data so that the performance of the system is reduced [6]. Of the above attacks, FDI attacks, which alter the data in the cyber layer, are more dangerous and more complex [7]. Based on where the attack occurs, FDI attacks can also be classified into actuator attacks [8, 9] and sensor attacks [10–13]. Detecting these attacks, which have an effect similar to the occurrence of a fault in the system, can prevent damage to it.

The ability to detect faults and cyber attacks can have a useful effect on how well the SCADA system responds to different conditions. The origins of a fault and of a cyber attack are different, but they have almost the same effect on the system. For example, a deviation in a sensor may be caused by a fault in it or by a cyber attack on its communication channel. Therefore, physical data alone cannot separate a fault from a cyber attack. A cyber attack will very probably have a particular effect on the network. It is therefore reasonable to study the network parameters as well as the physical data of the system [14–16].

In computer networks, any device that can send and receive information in the network is called a node. Also, to exchange information, large data are divided into smaller parts, and each of these parts is called a packet. The network parameters whose changes can be examined are the end-to-end delay, the jitter and the network throughput. The end-to-end delay is the time from when a packet is created at the sending node until it successfully reaches the receiving node. The variance of the end-to-end delay is called jitter. The network throughput is the rate of successful message delivery over the communication network, and its unit is bits per second (bps) [14].

This paper examines the possibility of detecting faults and cyber attacks and separating them from each other. For this purpose, the Beard fault detection filter (BFDF) is used to detect the anomalies that can occur in the system. The BFDF is a linear observer whose gain is obtained in such a way that, when a fault occurs, the residual responds in a specific and fixed direction. Therefore, in the residual space each fault has its own unique direction, and alignment of the residual with that direction identifies the type of fault in the system [17, 18]. A number of researchers have also used standard model-based fault detection algorithms, such as the unknown input observer (UIO), to detect cyber attacks [19–21]. Homaei [22] modeled an FDI attack in a power system and detected it using a Kalman filter and a χ^2 detector.

Examining the network parameters alongside the filter output adds a dimension to the residual space, because these two variables are independent of each other. In this new space it becomes possible to separate faults from cyber attacks.

Werth uses physical data and network information to separate faults from cyber attacks with machine learning algorithms. He does this by simulating different states of a system in the Omnet++ software [15]. Omnet++ is an extensible, library-based and component-based software package that is also used to build network simulations for cyber-physical systems, and it includes wireless and wired communication networks as well as different network topologies and protocols [23, 24]. Rønneberg also studied the effect of a cyber attack on network parameters by simulating two systems in Omnet++ and creating a cyber attack in them. He observed that network parameters such as the end-to-end delay, the jitter and the packet loss increased while the cyber attack was taking place [14].

In this paper a networked control system is introduced and the BFDF filter is designed for it. The system is then simulated in Omnet++ so that instantaneous access to the network parameters is also available. Finally, based on the filter output and the network parameters, the occurrence of a fault or a cyber attack is detected.

2. Materials and Methods

In this section the BFDF filter is first introduced and its details are examined. Then, in the case study section, a system is modeled and a BFDF filter is designed for it. Next, the network parameters are used to distinguish faults and cyber attacks from each other.

2.1 Beard fault detection filter (BFDF)

The state-space form of a discrete linear time-invariant system is defined as in Eq. (1).

$$\begin{aligned} x_{k+1} &= Ax_k + Bu_k + b_i\xi_{i_k} + \Delta A\,\alpha(x_k) + \Delta B\,\beta(u_k) \\ y_k &= Cx_k + I_j\eta_{j_k} \end{aligned} \tag{1}$$

The term $\Delta A\,\alpha(x_k) + \Delta B\,\beta(u_k)$ represents a fault in the system (deviation and perturbation in the system parameters). The terms $b_i\xi_{i_k}$ and $I_j\eta_{j_k}$ represent an anomaly (a fault or a cyber attack) in the actuators and the sensors of the system, respectively. Here $b_i \in \mathbb{R}^n$ is the i-th column of the matrix B, and $I_j \in \mathbb{R}^m$ is the unit vector indicating the anomaly in the j-th sensor.

Assuming the system is observable, the BFDF filter can be defined for it as in Eq. (2). Of course, if the system is not observable, the filter design process involves more details, which can be studied in [25–27].

$$\begin{aligned} z_{k+1} &= Gz_k + Dy_k + Bu_k \\ r_k &= y_k - Cz_k \\ G &= A - DC \;\rightarrow\; D = (A-G)C^{-1} \\ G &= \sigma_f I, \quad |\sigma_f| < 1 \end{aligned} \tag{2}$$

The matrix D is the observer gain and must be designed so that the residual has a unidirectional property in the presence of each specific anomaly. The state estimation error is $\varepsilon_k = x_k - z_k$. Now, if an anomaly is created in the actuators, the equation for the state estimation error and the residual become Eq. (3).

$$\begin{aligned} \varepsilon_{k+1} &= (A-DC)\varepsilon_k + b_i\xi_{i_k} \\ r_k &= C\varepsilon_k \end{aligned} \tag{3}$$

Therefore the residual in the output space changes in the direction of Cb_i. If a fault occurs in the system, it appears in the output space as a linear combination of ΔA and ΔB.

$$\begin{aligned} \varepsilon_{k+1} &= (A-DC)\varepsilon_k + \Delta A\,\alpha(x_k) + \Delta B\,\beta(u_k) \\ r_k &= C\varepsilon_k \end{aligned} \tag{4}$$

Also, if a deviation occurs in the sensors, the change in the residual is as in Eq. (5). In Eq. (5), d_j is the j-th column of the matrix D.

$$\begin{aligned} \varepsilon_{k+1} &= (A - DC)\varepsilon_k - d_j \eta_{j_k} \\ r_k &= C\varepsilon_k + I_j \eta_{j_k} \end{aligned} \tag{5}$$

An anomaly in the sensors and actuators can be caused by a fault in them or by the occurrence of a cyber attack. The BFDF filter detects only the occurrence of the anomaly and its location. Next, the network parameters, as parameters independent of the physical data, add a dimension to this residual space, and the separation of faults from cyber attacks is achieved in that space.

2.2 Case study

In this paper a fluid-flow system with two tanks was studied. This system has two pressure sensors and one actuator (a control valve). By modeling it and substituting the parameter values [15], its state-space form became Eq. (6).

$$\begin{aligned} p_{k+1} &= \begin{bmatrix} 0.9999 & 8.69\text{e}{-5} \\ 8.69\text{e}{-5} & 0.9999 \end{bmatrix} p_k + \begin{bmatrix} 6366 \\ 0.2769 \end{bmatrix} u_k \\ y_k &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} p_k \end{aligned} \tag{6}$$

If the pipe connecting the tanks breaks, the model of the system changes. When this fault occurs, the pressures of

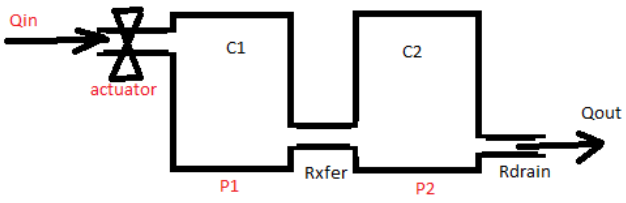


Fig. 1: Schematic of the two-tank fluid system

the tanks become independent of each other and the input affects only tank 1, and the state-space form of the system becomes Eq. (7).

$$\begin{bmatrix} p1_{k+1} \\ p2_{k+1} \end{bmatrix} = \begin{bmatrix} 0.9999 & 0 \\ 0 & 0.9999 \end{bmatrix} \begin{bmatrix} p1_k \\ p2_k \end{bmatrix} + \begin{bmatrix} 6366 \\ 0 \end{bmatrix} F_{in} \tag{7}$$

The state-space form of this system in the presence of anomalies in the actuator (f_a), in the sensors (f_s) and a fault in the system parameters (f_p) therefore becomes the following.

$$\begin{aligned} p_{k+1} = &\begin{bmatrix} 0.9999 & 8.69\text{e}{-5} \\ 8.69\text{e}{-5} & 0.9999 \end{bmatrix} p_k + \begin{bmatrix} 6366 \\ 0.2769 \end{bmatrix} u_k + \begin{bmatrix} 6366 \\ 0.2769 \end{bmatrix} f_{a_k} \\ &+ \begin{bmatrix} 0 & -8.69\text{e}{-5} \\ -8.69\text{e}{-5} & 0 \end{bmatrix} p_k f_{p_k} + \begin{bmatrix} 0 \\ -0.2769 \end{bmatrix} u_k f_{p_k} \\ y_k = &\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} p + \begin{bmatrix} 1 \\ 0 \end{bmatrix} f_{s1_k} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} f_{s2_k} \end{aligned} \tag{8}$$

In Eq. (8) the value of f_{p_k} is zero. If a fault occurs in the parameters of the system, its value becomes one and the system becomes as in Eq. (7). Under normal conditions f_{a_k}, f_{s1_k} and f_{s2_k} are also zero, and they take non-zero values at the time an anomaly occurs in the actuator and in sensors 1 and 2, respectively.

A PI controller was designed for this system. The control objective was to control the pressure of tank 2 so that it follows the reference signal (Fig. 2).

A filter like the one in Eq. (2) was designed for this system. If an anomaly occurs in the actuator, the residual in the

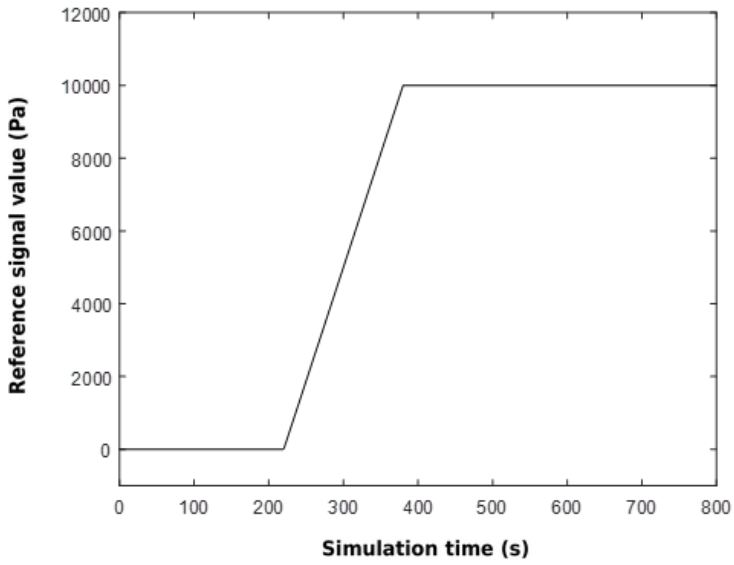


Fig. 2: Reference signal value for controlling the pressure of tank 2

output space changes in the direction of Cb, as in Eq. (3).

$$\sigma_f = 0.8 \;\rightarrow\; G = \begin{bmatrix} 0.8 & 0 \\ 0 & 0.8 \end{bmatrix}$$
$$A = \begin{bmatrix} 0.9999 & 8.69\text{e}{-5} \\ 8.69\text{e}{-5} & 0.9999 \end{bmatrix} \tag{9}$$
$$D = (A-G)C^{-1} \xrightarrow{C=I} D = \begin{bmatrix} 0.1999 & 8.69\text{e}{-5} \\ 8.69\text{e}{-5} & 0.1999 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad CB = \begin{bmatrix} 6366 \\ 0.2769 \end{bmatrix} \approx \begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{10}$$

In Eq. (2), σ_f is an arbitrary value of magnitude less than one, and in Eq. (9) its value was taken to be 0.8. The occurrence of a fault in the parameters of the system (breaking of the pipe connecting the tanks) appears in the output space as in Eq. (4), and according to Eq. (8) the values of ΔA and ΔB are obtained as in Eq. (11).

$$\Delta A = \begin{bmatrix} 0 & -8.69\text{e}{-5} \\ -8.69\text{e}{-5} & 0 \end{bmatrix}, \qquad \Delta B = \begin{bmatrix} 0 \\ -0.2769 \end{bmatrix} \tag{11}$$

Given that both ΔB and u_k have small values, $\Delta B\beta(u_k)$ (in Eq. (2) [sic]) does not have a large effect on the changes in the residuals. At the time the fault occurs, the increase in the controller error produces a large control signal, and this causes a large increase in the value of p_{1_k}. The quantity p_{1_k} also affects r_{2_k}. Therefore the output changes at the time this fault occurs are in the direction of $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. To obtain the directional relationship between the residual vector $r_k = \begin{bmatrix} r_{1_k} \\ r_{2_k} \end{bmatrix}$ and the anomaly signature vector (f), we use the correlation between them, as in Eq. (12) [28].

$$CORR_k = \frac{f^T r_k}{\|f\|_2 \, \|r_k\|_2} \tag{12}$$

The closer the correlation is to $+1$, the more the residual and the fault signature vector point in the same direction, and if they point in opposite directions the value is closer to -1. In any case, closeness to -1 or $+1$ means that the residual vector is aligned with the anomaly signature vector. Since the direction is what matters to us, we use its absolute value. We defined two correlations according to the signatures: one in the direction of r_{1_k} or $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, and the other in the direction of r_{2_k} or $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

$$CORR_{1_k} = \frac{|r_{1_k}|}{\|r_k\|_2}, \qquad CORR_{2_k} = \frac{|r_{2_k}|}{\|r_k\|_2} \tag{13}$$

To detect an anomaly in the sensors we also define two new correlations. If we assume that an anomaly occurs in both sensors at the same time, the equation for the state estimation error becomes Eq. (14). By substituting the values, the residual output equation becomes Eq. (15). In Eq. (16) we obtain the matrix Q in such a way that, after multiplying it by r_k, the effects of f_1 and f_2 are created in the directions of $\begin{bmatrix}1\\0.1\end{bmatrix}$ and $\begin{bmatrix}0.1\\1\end{bmatrix}$. Q was obtained according to Eq. (16).

$$\begin{aligned}
&\xrightarrow{(5)} e_{1_{k+1}} = \sigma_f e_{1_k} - d_{11}f_1 - d_{12}f_2 \rightarrow e_{1_k} \approx \frac{-d_{11}f_1 - d_{12}f_2}{1-\sigma_f} \\
&\xrightarrow{(5)} e_{2_{k+1}} = \sigma_f e_{2_k} - d_{21}f_1 - d_{22}f_2 \rightarrow e_{2_k} \approx \frac{-d_{21}f_1 - d_{22}f_2}{1-\sigma_f} \\
&\xrightarrow{(5)} r_{1_k} = \frac{-d_{11}f_1 - d_{12}f_2}{1-\sigma_f} + f_1 = \frac{(1-\sigma_f - d_{11})f_1 - d_{12}f_2}{1-\sigma_f} \\
&\xrightarrow{(5)} r_{2_k} = \frac{-d_{21}f_1 - d_{22}f_2}{1-\sigma_f} + f_2 = \frac{(1-\sigma_f - d_{11})f_2 - d_{21}f_1}{1-\sigma_f}
\end{aligned} \tag{14}$$

$$r_k = \begin{bmatrix}r_{1_k}\\r_{2_k}\end{bmatrix} = \begin{bmatrix}0.0005 & -0.0004\\-0.0004 & 0.0005\end{bmatrix}\begin{bmatrix}f_1\\f_2\end{bmatrix} \tag{15}$$

$$\begin{aligned}
&\begin{bmatrix}0.0005 & -0.0004\\-0.0004 & 0.0005\end{bmatrix}\begin{bmatrix}q_1\\q_2\end{bmatrix} = \begin{bmatrix}1\\0.1\end{bmatrix} \\
&\begin{bmatrix}0.0005 & -0.0004\\-0.0004 & 0.0005\end{bmatrix}\begin{bmatrix}q_3\\q_4\end{bmatrix} = \begin{bmatrix}0.1\\1\end{bmatrix} \\
&Q = \begin{bmatrix}q_1 & q_2\\q_3 & q_4\end{bmatrix} \rightarrow Q = \begin{bmatrix}\mathbf{4.4682} & \mathbf{3.9869}\\\mathbf{3.9869} & \mathbf{4.4682}\end{bmatrix}
\end{aligned} \tag{16}$$

We multiply the matrix Q by r_k, and, as in Eq. (17), two correlations are obtained based on $m_k = Qr_k$.

$$CORR_3(k) = \frac{|m_1(k)|}{\|m(k)\|_2}, \qquad CORR_4(k) = \frac{|m_2(k)|}{\|m(k)\|_2} \tag{17}$$

2.3 Adding network parameters to the BFDF filter data

The correlations defined in the previous section indicate the occurrence of anomalies in different parts of the system. These anomalies may be caused by a fault or by a cyber attack. Here we also examine the network information alongside the values of the correlations and the residual.

The network parameter given most attention in this paper is the network traffic. Given that the system is closed and has no incoming traffic, and that the number of connections between nodes and the normal traffic level of each node are known, an increase of the network traffic beyond a certain limit relative to the known value of each node can indicate an intrusion into the network and a cyber attack. The system introduced in the previous section was simulated in the Omnet++ software using the INET library.

The state-space form of the system was written in C++ and, as in Fig. 3, was placed in the plant section. The code of the controller and of the fault detection filter was also placed in the controller block. Part of the code is given in the appendix. The topology of the connection between the plant and the controller was also chosen as in Fig. 3. On one path the values of the sensors are sent to the controller, and on the other path the value of the control signal is sent to the plant. To make the internal parts of the plant and controller nodes clearer, the block diagram of the system is shown in Fig. 4.

In the attack scenarios it is assumed that the attacker already has access to the controllable excitation points needed to send packets through the network. During the attack, the attacker nodes, as in Fig. 3, begin to flood the communication network with packets and fill the access points, switches and router with packets. During a false data injection attack, the value of the transmitted data is changed on its way to the destination. The various anomaly states and the normal state, as in Fig. 5, were applied to the system in the simulation.

3. Simulation Results

To view the complete simulation codes in Omnet++, you can refer to this address.

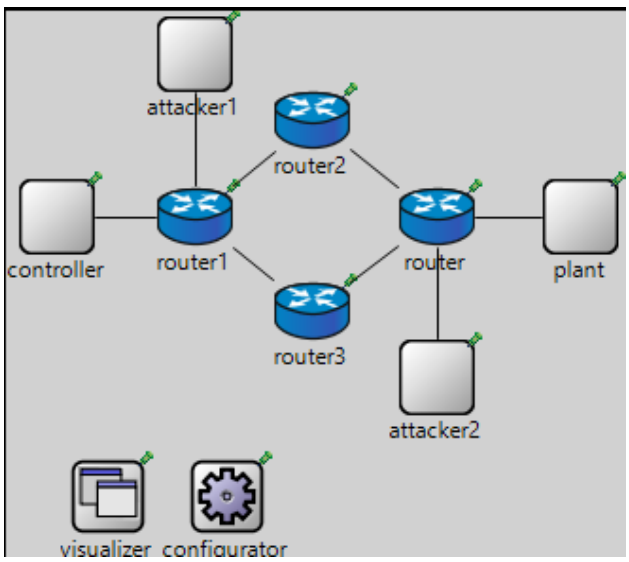


Fig. 3: Structure of the network designed in Omnet++

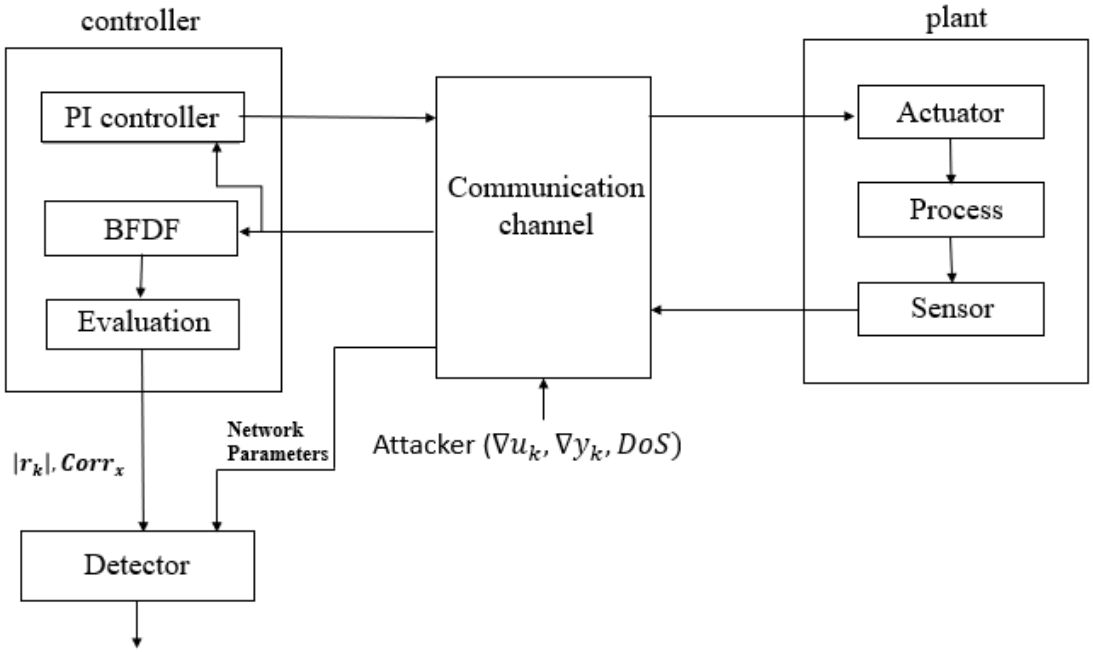


Fig. 4: Block diagram and structure of the system

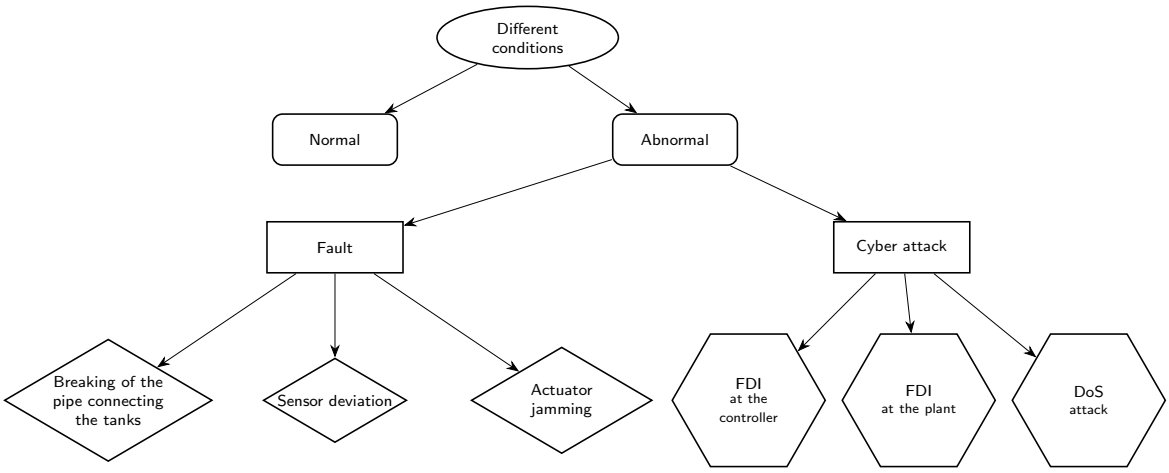


Fig. 5: Diagram of the different states of the system

The controller generates a control signal based on the value of the reference signal and the value of sensor 2, and sends it to the system. The system also sends the values of the sensors to the controller every 0.01 s using the UDP protocol. The length of each packet sent between nodes is 20 bytes. Therefore, given that the system is closed and the number of connections between nodes is known, the approximate traffic received by each node is known. Here, the traffic received by each node under normal conditions is about 16,000 bits per second, because each node receives 20 bytes of data every 0.01 s.

We take the four correlations obtained in the previous section as the criterion for separating the anomalies. The maximum of these four correlations gives us the type of the anomaly. The total simulation time is 800 seconds. In the time interval from 250 to 500 seconds, the anomalies shown in Fig. 5 were created in several separate simulations. The results of these simulations are given in Table 1.

Table 1: Correlation values at the time of occurrence of each anomaly

	$Corr_1$	$Corr_2$	$Corr_3$	$Corr_4$
f_u	1	0.02	0.74	0.66
f_p	0	1	0.66	0.74
f_{s1}	0.75	0.65	1	0.1
f_{s2}	0.65	0.75	0.1	1

As an example, when a cyber attack occurs in the communication channel of sensor 1, the residual magnitude, the correlations and the network parameters are as in Figs. 6 to 12. In the figures, correlation 3 is larger than the other correlations and its value has become one. An anomaly in sensor 1 is therefore detected, and given the network traffic and the fact that it is greater than the expected value, the occurrence of a cyber attack in the channel of sensor 1 is detected.

It is clear that the value of the correlation at other times is not important, because at those times the residual has a value close to zero and no anomaly has been detected. The magnitude of the residual at the time of the anomaly was about 330, and at other times it is close to zero.

In Table 1, f_u, f_{s1} and f_{s2} are, respectively, the anomaly (attack or fault) in the actuator, in sensor 1 and in sensor 2, and f_p is a fault in the system. The values of the correlations at the time these anomalies occur are written in front of them.

If a deviation occurs in the sensors or in the actuator, it can be caused by a fault in them or by the occurrence of a cyber attack in their channel. In both cases, the correlations determine the location where the anomaly occurs, and the network parameters determine the type of the anomaly.

If a DoS attack occurs in the communication channel of the actuator, the value of the control signal stays constant and appears like a fault in the actuator. If it occurs in the communication channel of the sensors, their values stay constant and do not change with changes in the control signal. At the time of the DoS attack, the network traffic reaches such a level that new data do not reach the destination or arrive with a large time gap.

One way to detect a DoS attack is to examine the network traffic. If the network traffic becomes, for example, twice

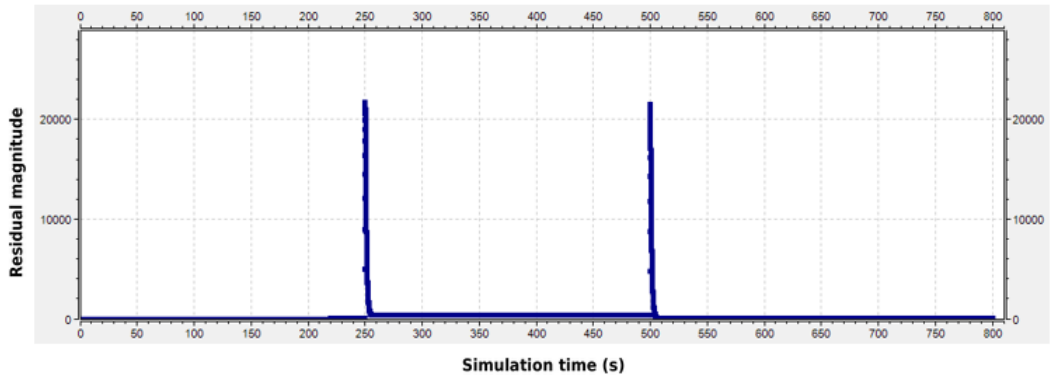


Fig. 6: Residual magnitude at the time of the attack on sensor 1 during the simulation time (s)

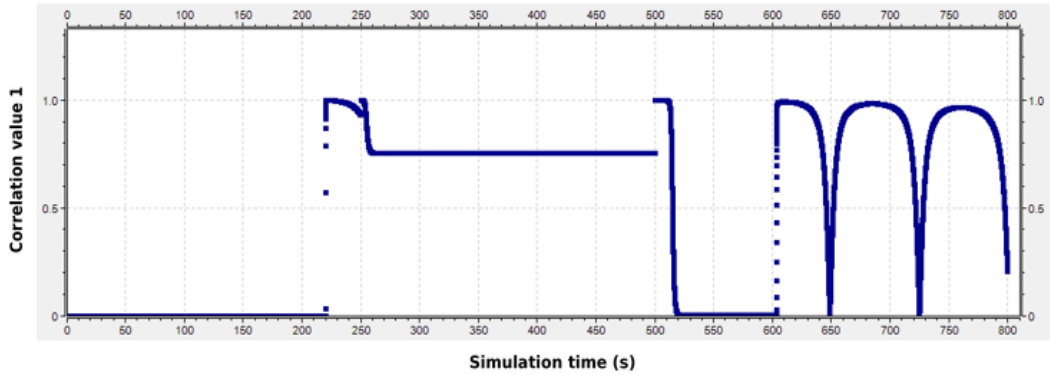


Fig. 7: Correlation value 1: signature of the anomaly in the actuator versus simulation time (s)

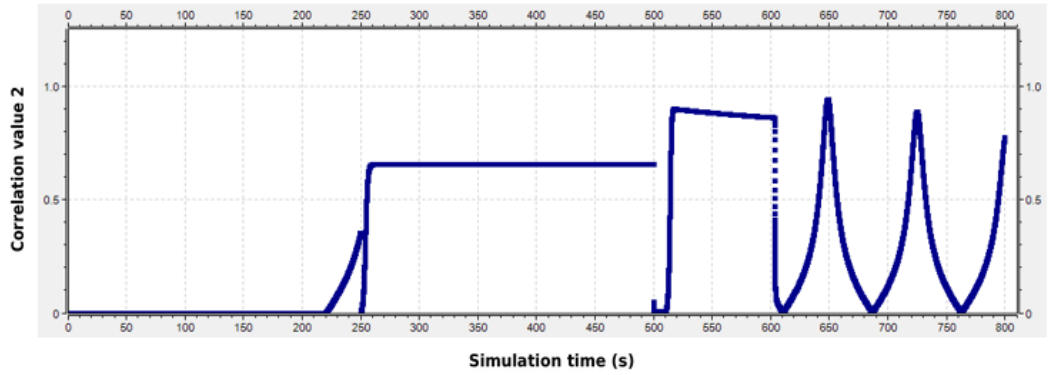


Fig. 8: Correlation value 2: signature of the fault in the system parameters versus simulation time (s)

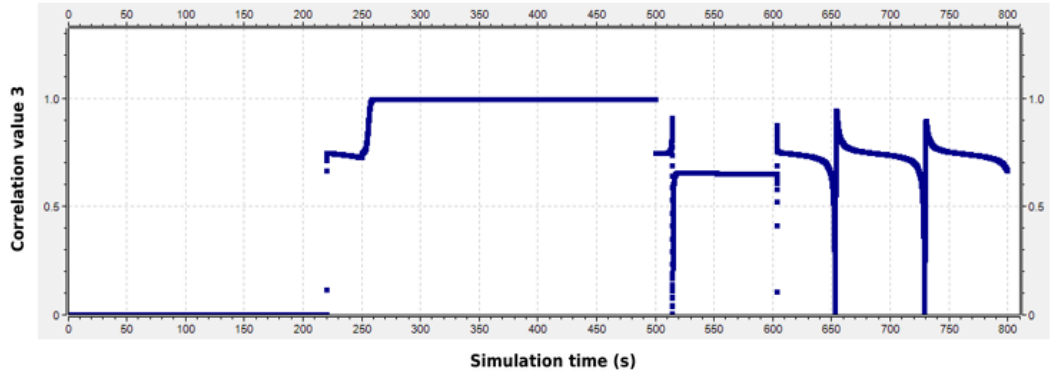


Fig. 9: Correlation value 3: signature of the anomaly in sensor 1 versus simulation time (s)

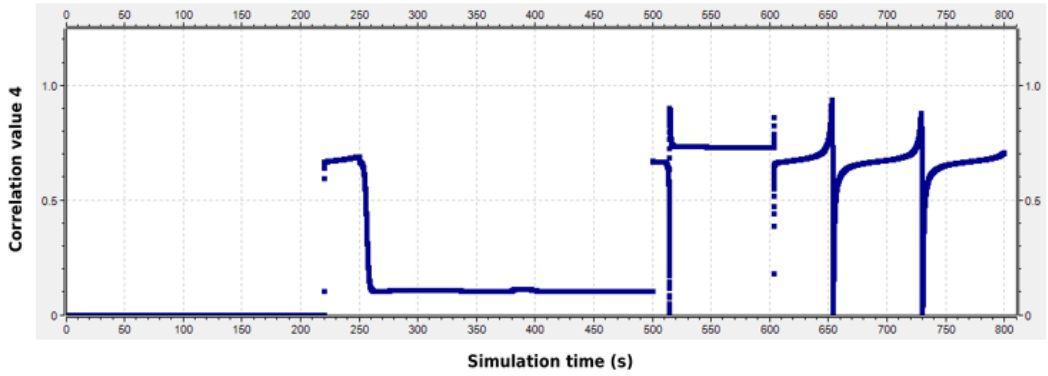


Fig. 10: Correlation value 4: signature of the anomaly in sensor 2 versus simulation time (s)

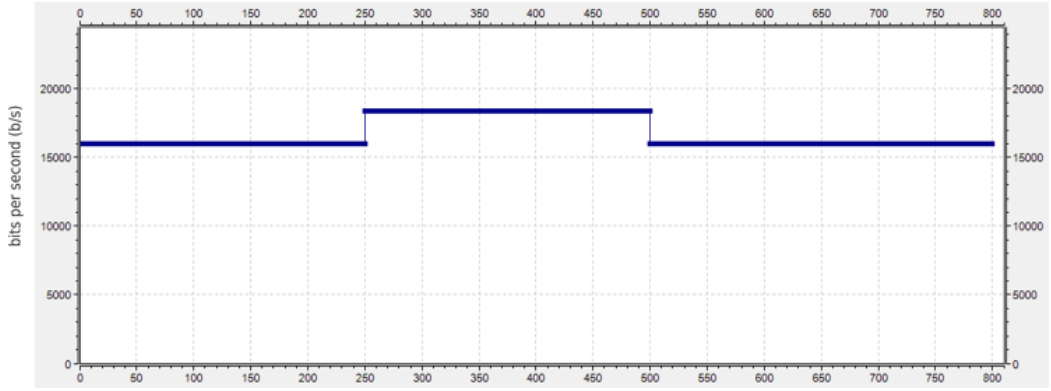


Fig. 11: Network traffic (bits per second) versus simulation time (s)

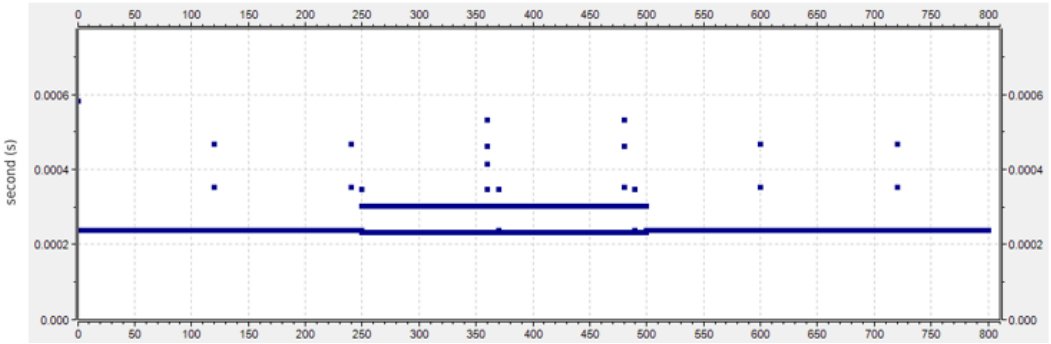


Fig. 12: End-to-end delay versus simulation time (s)

its normal value, a DoS attack is detected.

Another way was also considered for detecting a DoS attack in the channel of the sensors. We define a variable with the value zero. In each sample we compare the current and previous values of the control signal, and likewise those of the sensors. If the value of the control signal changes but the values of the sensors stay constant, we increase the variable by one. If, for example, this happens for 50 consecutive samples and the variable reaches 50, a DoS attack is detected. Since the packet send and receive time was taken to be 0.01 s, 50 samples take about 0.5 s, which is a suitable detection time.

If a deviation is created in both sensors at the same time, its effect shows in the magnitude of the residual, but none of the correlations becomes 1. To find out how much deviation there is in each sensor, one can take the ratio of correlations 3 and 4. In this way one can tell in which sensor the deviation is larger and how many times larger it is than in the other.

Therefore, according to the results obtained, an algorithm like the one in Fig. 13 can be presented for detecting and isolating faults and cyber attacks. The number 50 in the flowchart relates to the detection of the DoS attack explained above. Since, when an anomaly occurs, the corresponding correlation will have a value close to 1, a value greater than 0.98 was used in the flowchart for each anomaly to be detected.

4. Conclusion

The main goal of this paper was to examine some well-known cyber attacks and to evaluate how they can be distinguished from the common faults that can occur in a system. To understand the state of a networked control system better, this paper used network parameters in addition to physical data. For this purpose, a networked control system was modeled in which the controller and the system communicated with each other through Internet protocols. To detect the anomalies of the system, both faults and cyber attacks, a model-based method, namely the BFDF filter, was designed for the system.

Each anomaly in the residual space produced by the filter has its own unique direction and can be separated from the others. Next, to detect whether an anomaly is caused by a fault or by a cyber attack, the use of network parameters was proposed. The system was simulated in Omnet++ software and various faults and cyber attacks were applied to it. Finally, based on the results obtained, an algorithm for detecting faults and cyber attacks was written for the system.

Suggestions

For future work, one can point to the design of a new type of intrusion detection system that is more specific to SCADA systems. To separate faults from cyber attacks, one can also use other model-based methods and the simulation of more complex faults and cyber attacks, with the help of network parameters for better detection of them. One can also use machine learning classification methods in which network parameter data are added to the dataset.

Appendices

The C++ code of the system is as follows.

```
void Plant::RealPlant(double u)
{
  p1 = A[0][0] * p1 + A[0][1] * p2 + B[0] * u;
  p2 = A[1][0] * p1 + A[1][1] * p2 + B[1] * u;
}
```

The code of the PI controller is as follows.

```
void Controller::PI_Controller(double y)
{
  referenceValue = ReferenceSignal();
  controlError = referenceValue - y;
  Integral = Integral + Ts * controlError;
  controlSignal = Kp * controlError + Ki * Integral;

  if (controlSignal < 0) controlSignal = 0;

  emit(referenceValueSignal, referenceValue);
  emit(errorValueSignal, controlError);
}
```

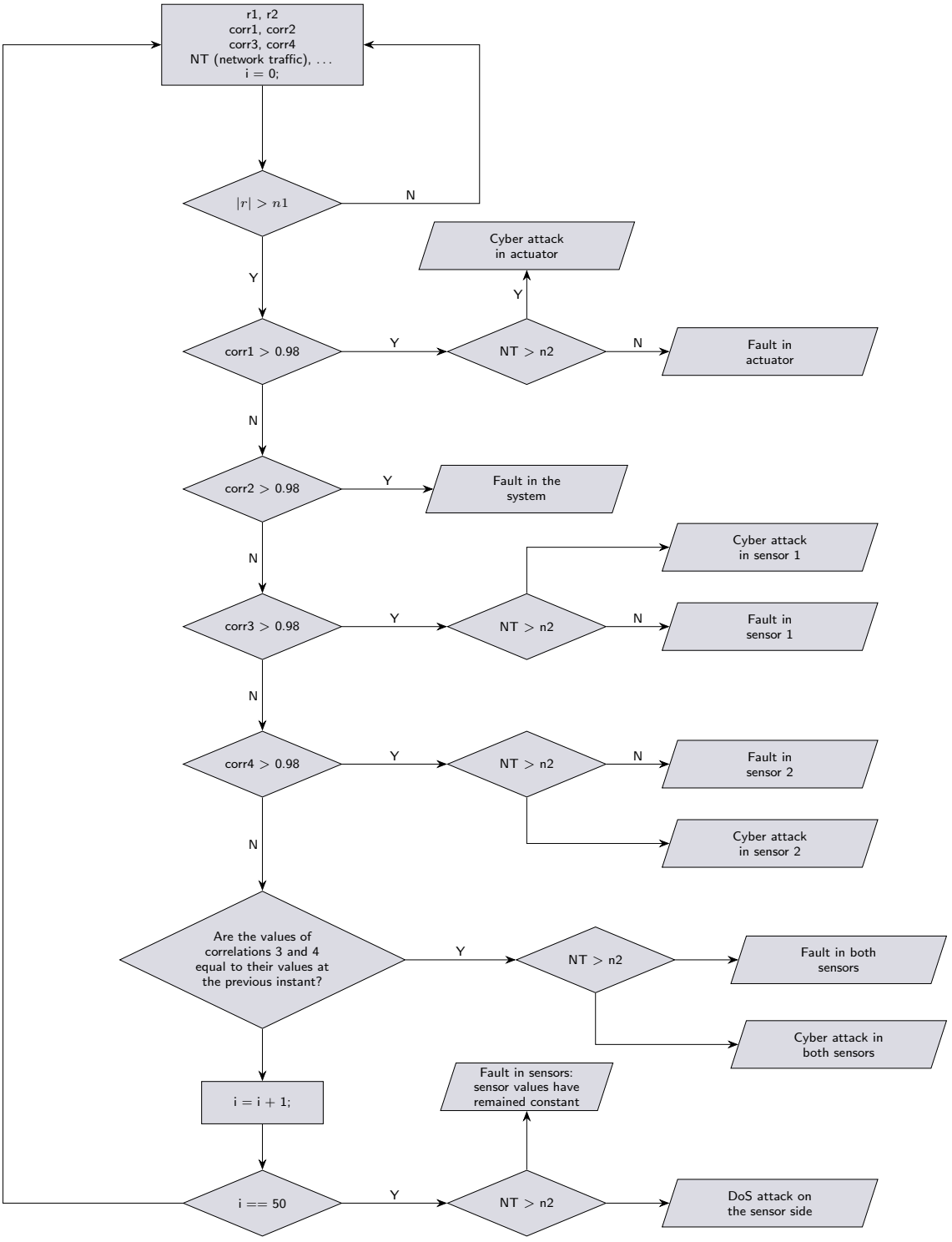


Fig. 13: Algorithm for detecting and isolating faults and cyber attacks, assuming the network parameters are available, using the Beard fault detection filter

The C++ code of the fault detection filter part, as well as the computation of the correlations, was also written as follows.

```
const double G[2][2] = {{0.8, 0},
                        {0, 0.8}};
const double D[2][2] = {{0.1999, 8.69e-5},
                        {8.69e-5, 0.1999}};
const double Q[2][2] = {{4.4682, 3.9869},
                        {3.9869, 4.4682}};

void Controller::Plant_Model()
{
  R1 = (p1 - z1);
  R2 = (p2 - z2);

  m1 = (Q[0][0] * R1 + Q[0][1] * R2);
  m2 = (Q[1][1] * R2 + Q[1][0] * R1);

  z1 = G[0][0] * z1 + G[0][1] * z2 + D[0][0] * p1 + D[0][1]
       * p2 + Bb[0] * u[0];
  z2 = G[1][0] * z1 + G[1][1] * z2 + D[1][0] * p1 + D[1][1]
       * p2 + Bb[1] * u[0];
}

void Controller::Evaluation()
{
  norm_R = sqrt(R1 * R1 + R2 * R2);
  norm_m = sqrt(m1 * m1 + m2 * m2);

  corr1 = abs(R1) / (norm_R + 0.001);
  corr2 = abs(R2) / (norm_R + 0.001);
  corr3 = abs(m1) / (norm_m + 0.001);
  corr4 = abs(m2) / (norm_m + 0.001);
}
```

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